

Cost Estimation Guide

This document provides detailed cost estimates for running the Audit Evidence Evaluation Pipeline, broken down by **Phase 1: Extraction** (cached) and **Phase 2: Evaluation** (not cached).

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Overview

The pipeline processes audit evidence in two distinct phases with different cost profiles:

Phase	Stage	Service Used	Cached?	Cost Profile
Phase 1: Extraction	Stage 2	Document Intelligence + GPT-4.1/5.1 Vision	[YES] Yes	~\$0.05-\$0.07/slide
Phase 2: Evaluation	Stage 4	GPT-4.1/5.1 (Text Only)	[NO] No	~\$0.016/element

Key Insight: On re-runs with the same source documents, you only pay for Phase 2 (evaluation), resulting in **83-87% cost savings**.

Phase 1: Extraction Costs (Cached)

Phase 1 extracts text and data from PowerPoint slides using Document Intelligence for OCR and GPT Vision for interpretation.

Azure Document Intelligence

Model	Price per Page	Use Case
prebuilt-layout	\$0.01	OCR for embedded images in slides

Azure OpenAI GPT-4.1 / GPT-5.1 Vision

Used to interpret slide images and extract structured audit data.

Model	Input (per 1K tokens)	Output (per 1K tokens)
GPT-4.1	\$0.002	\$0.008
GPT-5.1	\$0.00125	\$0.01

Token Estimation for Slide Extraction

Based on actual pipeline runs:

Component	Tokens	Notes
Image tokens (per slide)	~1,100	768x768 low-detail mode
System prompt	~500	Extraction instructions
Output (structured JSON)	~800-1,500	Varies by content density

Estimated cost per slide (GPT-4.1 Vision): - Input: ~1,600 tokens × \$0.002/1K = ~\$0.0032 - Output: ~1,200 tokens × \$0.008/1K = ~\$0.0096 - **Total GPT cost per slide: ~\$0.013** - **Total with DI OCR: ~\$0.023/slide** (if slide has embedded images)

For slides with multiple embedded images requiring OCR: - **Estimated: \$0.05-\$0.07 per slide**

Phase 2: Evaluation Costs (Not Cached)

Phase 2 evaluates extracted elements against PI requirements using GPT text models (no vision needed).

Azure OpenAI GPT-4.1 / GPT-5.1 (Text Only)

Model	Input (per 1K tokens)	Output (per 1K tokens)
GPT-4.1	\$0.002	\$0.008
GPT-5.1	\$0.00125	\$0.01

Token Estimation for Element Evaluation

Component	Tokens	Notes
System prompt	~800	Evaluation rubric and criteria
Element context	~500-1,000	Extracted text, tables, metadata
Output (evaluation JSON)	~300-500	Score, reasoning, recommendations

Estimated cost per element (GPT-4.1): - Input: ~1,200 tokens × \$0.002/1K = ~\$0.0024 - Output: ~400 tokens × \$0.008/1K = ~\$0.0032 - **Total per element: ~\$0.006**

With overhead for context and retries: - **Practical estimate: ~\$0.016 per element**

Model Comparison

GPT-4.1 vs GPT-5.1

Factor	GPT-4.1	GPT-5.1
Availability	Generally Available	Check region availability
Input Cost	\$0.002/1K tokens	\$0.00125/1K tokens
Output Cost	\$0.008/1K tokens	\$0.01/1K tokens
Vision Support	[YES] Yes	[YES] Yes
Recommended For	Production workloads	Latest capabilities

Note: GPT-5.1 pricing may vary. Always check the [Azure OpenAI Pricing Page](#) for current rates.

Total Cost Examples

Example: 69-Slide Presentation with 54 Evidence Elements

First Run (No Cache)

Phase	Items	Cost per Item	Total
Phase 1: Extraction	69 slides	~\$0.063/slide	~\$4.35
Phase 2: Evaluation	54 elements	~\$0.016/element	~\$0.86
Total First Run			~\$5.21

Subsequent Runs (With Cache)

Phase	Items	Cost per Item	Total
Phase 1: Extraction	69 slides	\$0.00 (cached)	\$0.00
Phase 2: Evaluation	54 elements	~\$0.016/element	~\$0.86
Total Cached Run			~\$0.86

Savings: ~\$4.35 (83% reduction)

Scaling Examples

Document Size	Elements	First Run	Cached Re-run	Savings
Small (20 slides, 15 elements)	15	~\$1.50	~\$0.24	84%
Medium (50 slides, 40 elements)	40	~\$3.79	~\$0.64	83%
Large (100 slides, 80 elements)	80	~\$7.58	~\$1.28	83%
Extra Large (200 slides, 150 elements)	150	~\$14.99	~\$2.40	84%

Caching Benefits

What Gets Cached (Phase 1)

The extraction cache stores: - [YES] Slide image renderings - [YES] Document Intelligence OCR results - [YES] GPT Vision extraction outputs - [YES] Structured JSON for each slide

What Doesn't Get Cached (Phase 2)

Evaluation results are regenerated each run because: - Evaluation criteria may change - Element scores need to reflect current requirements - Fresh analysis ensures accuracy

Cache Storage Options

Option	Storage	Best For
Azure Blob Storage	Cloud-based, persistent	Production, team sharing
Local File Cache	.extraction_cache/ folder	Development, testing

Re-run Cost Breakdown

When you re-run the pipeline with cached extraction:

```
First Run: █ Phase 1: Extraction █████ $4.35 (PAID) █ Phase 2: Evaluation █████ $0.86 (PAID) █████
Total: $5.21 Cached Re-run: █ Phase 1: Extraction █████ $0.00 (CACHED ✓) █ Phase 2: Evaluation █████
$0.86 (PAID) █████ Total: $0.86 (83% savings!)
```

Cost Optimization Tips

1. Leverage Caching

- Use Azure Blob Storage for persistent caching across sessions
- Only re-extract when source documents actually change
- Use --skip-cache flag sparingly (forces full re-extraction)

2. Batch Processing

- Process multiple documents in a single session
- Extraction cache is reusable across pipeline runs

3. Model Selection

- GPT-4.1 offers excellent price/performance for most use cases
- Consider GPT-5.1 for complex documents requiring latest capabilities

4. Image Optimization

- Use low detail mode for slide rendering (768x768)
- Reduces token count while maintaining accuracy

5. Monitor Usage

- Track Azure OpenAI token consumption in Azure Portal
- Set up cost alerts for unexpected spikes

References

- [Azure OpenAI Pricing](#)
- [Azure Document Intelligence Pricing](#)
- [GPT-4.1 Vision Token Calculation](#)
- [Azure Blob Storage Pricing](#)